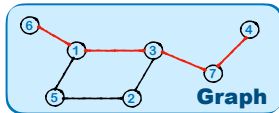


Query: The task is to determine the shortest path between two nodes. Here is an undirected graph containing nodes from 1 to 7. The edges are: (1, 3), (1, 6), (1, 5), (3, 2), (3, 7), (5, 2), (7, 4). Question: What is the shortest path between node 6 and node 4?

Ground Truth: [6, 1, 3, 7, 4]

Question



We can use Dijkstra's algorithm to find the shortest path.

Let's start by initializing the distances from node 4 to all other nodes and setting the distance to the starting node (node 4) to 0. The distance to all other nodes is initially set to infinity.

1. Initialize distances:
 - Distance to 4: 0
 - Distance to 7: ∞
2. Select the node with the smallest distance (node 4) and update the distances to its neighbors:
 - Node 4 has no neighbors.
8. Select the next smallest distance node (node 7), update the distances to its neighbors:

Since we have reached node 4 and the distance to it is 5, we can now trace back the path:

- From node 7 to node 4: distance is 2 (edge (7, 4))

The shortest path from node 6 to node 4 is: [6, 5, 2, 3, 7, 4]

Therefore, the final answer is: $\boxed{[6, 5, 2, 3, 7, 4]}$.

Qwen2.5-3B-Instruct

To determine the shortest path between node 6 and node 4, we will analyze the given graph step by step.

First, let's list the nodes and their connections:

- Node 1 is connected to nodes 3, 6, and 5.

We need to find the path from node 6 to node 4. Let's start from node 6 and see which nodes we can reach next.

1. Node 6 is connected to node 1.
2. Node 1 is connected to nodes 3, 6, and 5.
3. Since we are already at node 6, we can try to reach other nodes from here. Node 1 is connected to node 3, so let's see if we can get to node 4 from there.

Now, let's look at the connections from node 3:

Now, let's look at the connections from node 7:

1. Node 7 is connected to nodes 3 and 4.
2. Since we are already at node 7, we can reach node 4 directly from here.

So, the path from node 6 to node 4 is:
[6 to 1 to 3 to 7 to 4]

Therefore, the final answer is: $\boxed{[6, 1, 3, 7, 4]}$.

G1-Zero-3B

To find the shortest path between node 6 and node 4 in the given undirected graph, we can use a graph traversal method such as Breadth-First Search (BFS).

Let's represent the graph as an adjacency list for easier visualization: - Node 1: connected to 3, 6, 5

Starting from node 6, we will perform a BFS to find the shortest path to node 4.

1. Initialize the queue with the starting node 6.
2. Keep track of visited nodes to avoid revisiting.
3. Track the path by backtracking from the end node to the start node.

Let's execute the BFS:

- Start at node 6.
- Explore neighbors of 6: the only neighbor is 1.
- From node 1, explore its neighbors: 3, 5, 6 (but 6 is already visited).....
- From node 7, explore its neighbors: 4 (this is the target node).

Thus, the shortest path found is: 6 -> 1 -> 3 -> 7 -> 4.

Therefore, the final answer is: $\boxed{[6, 1, 3, 7, 4]}$.

G1-3B

Figure 2: An intuitive illustration of the differences in solution strategies employed by Qwen2.5-3B-Instruct, G1-Zero-3B, and G1-3B for a shortest path problem.